Building Deployable and Pragmatic Robots from Human Priors

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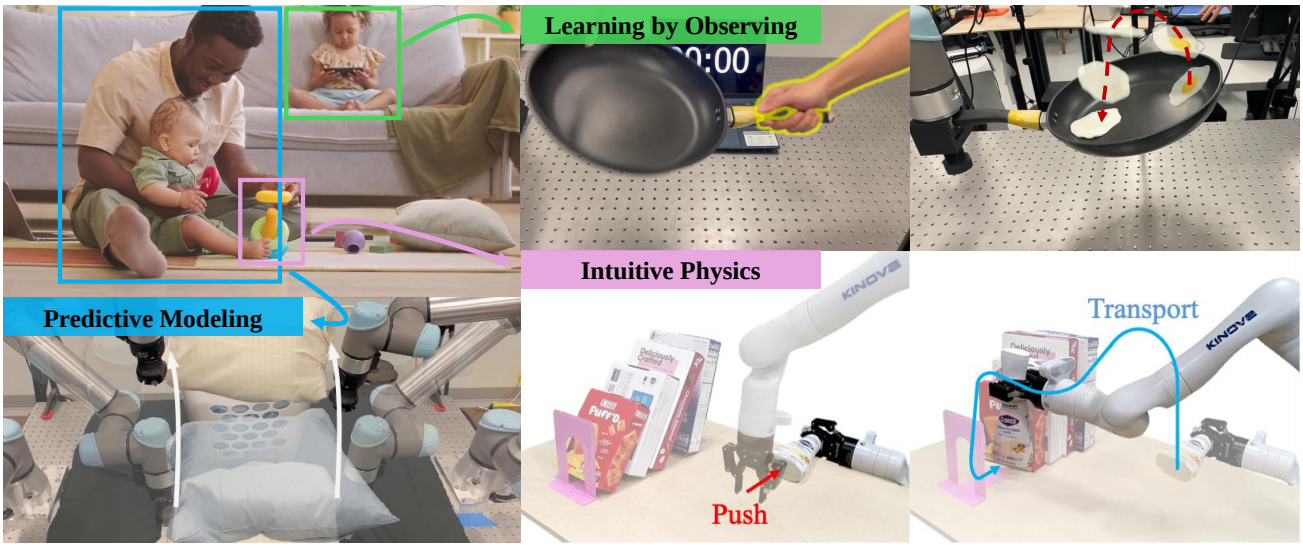


Fig. 1: **Building Deployable and Pragmatic Robots from Human Priors.** Humans possess **innate cognitive models (or *human priors*)** that enable intuitive reasoning, such as predicting physical interactions, anticipating future states during manipulation, and generalizing affordances across tools. Building on these insights, I develop deployable and pragmatic robots from various human priors for real-world tasks.

I. Introduction

My research goal is to build deployable and pragmatic robots that can rapidly acquire reasoning and interaction capabilities, much like how humans develop skills. Robots hold transformative potential for healthcare, logistics, and domestic assistance, yet their real-world deployment remains confined to controlled industrial settings. While humans effortlessly adapt to unstructured tasks—stowing objects in cluttered shelves, sorting deformable laundry with both arms, or improvising tool use—robots falter due to rigid, data-hungry learning paradigms. This gap persists because robots lack the **innate cognitive models (or *human priors*)** that enable intuitive reasoning, such as predicting physical interactions, anticipating future states during manipulation, and generalizing affordances across tools.

My research bridges this gap by **embedding human priors as inductive biases into robot learning**, enabling robots to reason, predict, and adapt with human-like efficiency. To achieve this, I develop a framework inspired by three core human priors: intuitive physics for object interactions, predictive modeling for decision-making, and tool affordance abstraction. I built upon graph-based neural dynamics models to develop a framework for stowing tasks, enabling robots to reason about multi-object interactions in cluttered environments [7] (Best Paper Finalist, CoRL 2023). Beyond manipulation, I extended these ideas to human-in-the-loop systems, enabling robots to leverage human behavior priors for adapting to diverse behaviors, improving traffic flow, and enhancing intersection navigation [1, 2, 4, 5]. For coordinated bimanual manipulation, such as laundry cleanup, I advanced state-of-the-art behavior cloning methods by enabling robots to anticipate object movement and translate predictions into synchronized motions [8]. Additionally, my research explores how robots can learn from human behaviors for safe and efficient human-robot collaboration [6, 3]. Furthermore, I enable robots to acquire extreme dynamic manipulation capabilities—such as using a pan to flip food—by using tools as the common interface from human play data. Together, these priors empower robots to generalize from minimal demonstrations, adapt to dynamically coordinated multi-agent scenes, and transfer skills across different embodiments, transforming them from scripted machines into intelligent agents that reason, predict, and generalize with human-like adaptability.

II. Previous Research

Intuitive Physics: Modeling Multi-Object Dynamics. Effective robotic manipulation in cluttered environments requires reasoning about multi-object interactions. Traditional methods rely on manually defined dynamics models or large labeled datasets, limiting scalability. **Since explicit**

contact modeling is challenging, I developed a graph neural dynamic model that predicts object interactions within the behavior primitive parameter space, enabling generalization across diverse object configurations [7].

My framework leverages GNN to predict object movements given robot actions. Instead of explicitly modeling contact forces and friction, it learns implicit interaction priors from off-policy data, improving data efficiency and generalization. Unlike reinforcement learning approaches, my method provides a structured action space using behavior primitives—such as pushing, sweeping, and transporting—simplifying long-horizon planning.

Our model enables zero-shot sim-to-real transfer, allowing robots trained in simulation to handle real-world clutter with varying object properties. By structuring manipulation around higher-level primitives, it enhances planning efficiency and enables complex, long-horizon task execution with minimal supervision. Results demonstrate strong generalization across diverse shelf configurations and object types.

Predictive Modeling for Coordinated Manipulation.

Effective bimanual coordination is essential for tasks like sorting laundry, picking from cluttered shelves, and bimanual fruit holding. Humans naturally anticipate how their actions will change an object's state, enabling adaptive and precise manipulation. However, robotics lacks this predictive capability, making it difficult to model object movement, predict future states, and generate coordinated actions. Traditional imitation learning struggles with multimodal goals, unseen object interactions, and unreliable state prediction, often leading to failures like object drops and improper force distribution. **To address these challenges, we introduce a prediction-aided imitation learning framework that enables robots to anticipate future object states, improving stability and task success [8].**

Our key insight is that explicitly modeling object movement improves bimanual coordination. Inspired by human strategies, we disentangle task-related state transitions from inverse dynamics modeling, allowing the robot to plan actions more effectively. Using a demonstration dataset, we train a diffusion model to predict future object states from past observations and an inverse dynamics model to compute actions that achieve these states. Unlike conventional approaches that treat policy learning as a black box, our framework separates state prediction from action generation, enhancing interpretability and enabling proactive failure recovery.

Evaluations across diverse simulation and real-world setups—including multimodal goal configurations, deformable objects, and multi-object interactions—show that our method consistently outperforms state-of-the-art state-to-action policies. By explicitly modeling future states, the robot anticipates and mitigates failures, maintains stability across control modes, and synthesizes a broader range of behaviors than those in the demonstration dataset. Our framework advances coordinated robotic manipulation, paving the way for scalable deployment in industrial, assistive, and household robotics.

Tool Interface Abstraction: Learning from Human Play.

Robotic tool use is crucial for complex tasks, but traditional data collection—via teleoperation or kinesthetic teaching—is inefficient, requiring direct robot access and expert operators. These methods struggle with dynamic tool-use tasks like dynamic object flipping, bottle balancing, or soccer striking due to control delays and restrictive action spaces. In contrast, human play offers a scalable way to collect diverse demonstrations without specialized hardware. However, leveraging human play presents challenges, including embodiment mismatches and viewpoint variations.

We propose a framework that transfers human play data into deployable robot policies by using the tool as a common interface. Our method employs 3D reconstruction and Gaussian splatting for novel view augmentation, segmentation models to filter embodiment-specific information, and tool-centric representation learning to ensure policy robustness under varying robot bases and viewpoints. Our method allows robots to learn tool-use skills from raw human play without manual intervention.

Empirical results show our framework reduces data collection time by 77%, improves success rates by 71% over teleoperation-based policies, and enables tasks like dynamic flipping and fine-grained balancing that prior methods fail to solve. Moreover, it generalizes across robot embodiments, camera viewpoints, and object variations, making it scalable, hardware-efficient, and practical for real-world deployment. By eliminating the need for expensive data collection devices, our method significantly lowers the barrier for training robots in unstructured tool-use environments.

III. Research Vision for Future Work

My future research will focus on advancing robot learning for generalization and scalable deployment for manipulation. While humans effortlessly adapt skills across environments, robots still struggle to generalize. The key challenge is enabling robots to **flexibly compose learned models and policies** to handle unseen tasks.

Building upon my prior work in (1) intuitive physics, (2) predictive reasoning, and (3) tool affordance learning, I propose a **structured hybrid learning framework** that dynamically selects between model-based and policy-based learning depending on representation complexity. Our approach will allow robots to generalize better, learn efficiently from minimal demonstrations, and adapt to new environments. While this serves as my primary research direction, I remain open to exploring other promising approaches that contribute to the broader goal of scalable, adaptive robot learning.

Learning Structured World Models. When policy learning is difficult due to limited on-policy data or long-horizon task nature, a model-based approach enables robots to simulate and plan before execution, reducing reliance on direct trajectory learning. For example, predicting how a teaspoon moves when tapping the edge of a cup to remove excess tea leaves is more effectively handled by learning a structured physics model rather than relying on imitation or reinforcement learning. A structured world model can condition on task-relevant physics

priors and scene embeddings to generalize better across spoon shapes, cup sizes, and tapping forces—with minimal demonstrations. My prior work in graph-based interaction modeling has demonstrated that learning object dynamics through structured priors significantly improves zero-shot sim-to-real transfer and enhances long-horizon planning in cluttered environments [7].

Direct Skill Acquisition Through Policy Learning. When model learning is challenging due to high system complexity or hard-to-modeled state space, directly learning policies bypasses explicit modeling. For instance, grasping a delicate tea bag involves handling deformable materials with unknown physics. Here, a direct policy learning is more effective than constructing an explicit model. My prior research in tool-use policy transfer has shown that robots can derive policies directly from human play, acquiring skills without hardware-specific constraints.

Composable Model and Policy Learning Instead of training monolithic policies or models for each task, my research focuses on a modular, composable learning approach, where object dynamics, affordances, and skill representations remain independent yet combinable. Robots will learn to compose task-agnostic skill primitives—such as grasping, tilting, and pouring—alongside world models of object interactions. This enables adaptation to new kettles, cups, and tea bags without retraining, even in multi-arm settings.

To achieve this, I will integrate energy-based models, foundation models, and graph neural networks. Energy-based models, such as diffusion models, offer composability, while graph neural networks capture spatial relationships between objects like the kettle, cup, and hand pose. **By combining learned models and policies, the robot can rapidly adapt to new environments with minimal fine-tuning, reducing reliance on large-scale retraining.**

The research will enable lifelong learning agents that balance model-based and model-free learning, making robots more scalable, adaptive, and generalizable across industries such as manufacturing, healthcare, and assistive robotics.

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